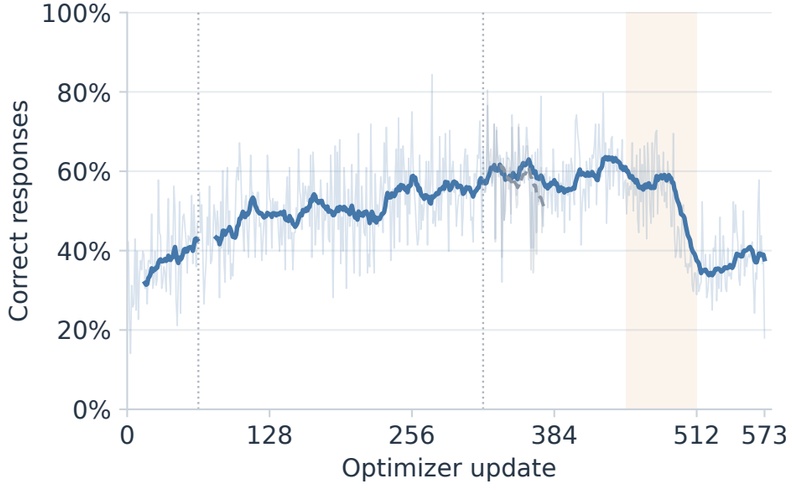
Snowball RL: optimizer and rollout diagnostics from the start

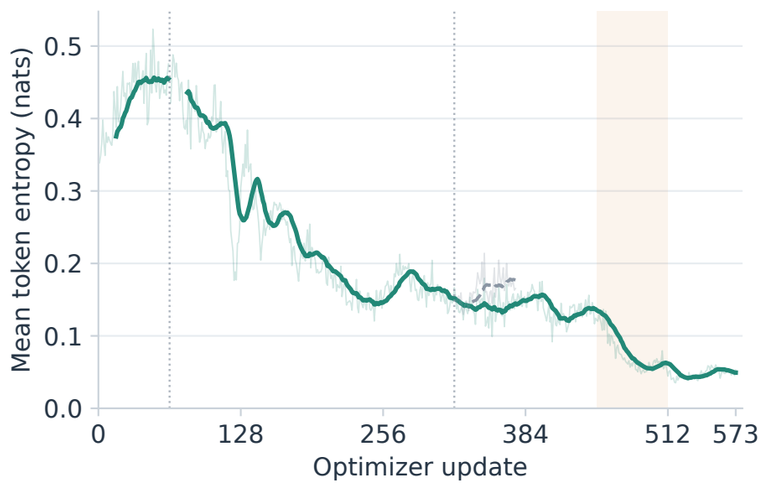
Steps 0–573 • faint: each update • colored line: trailing 16-update mean • gray dashed: discarded branch

Dotted vertical lines: scale-up at 64 and restart at 320 • shaded interval: 448–512

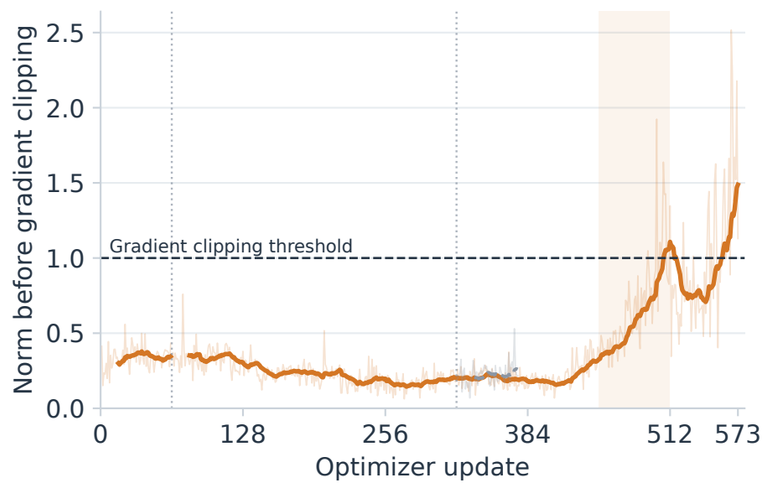
A Training success rate



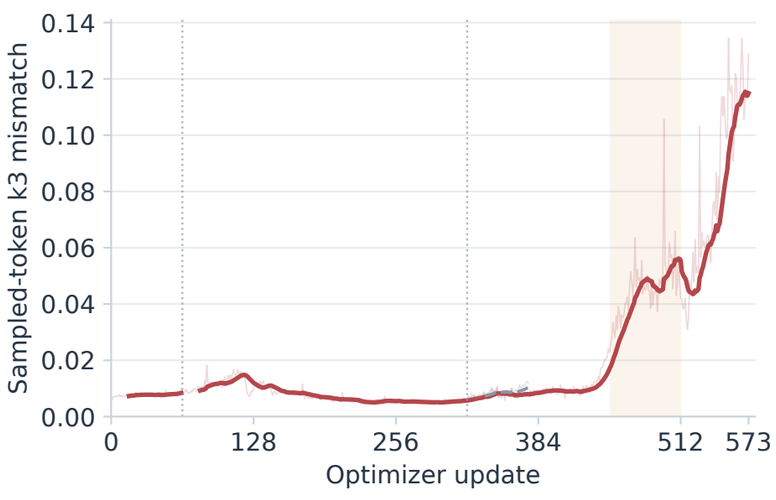
B Policy entropy



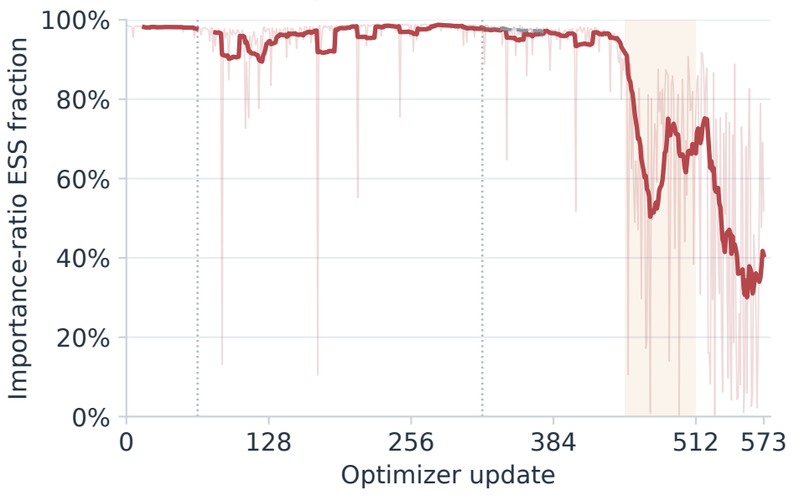
C Gradient norm



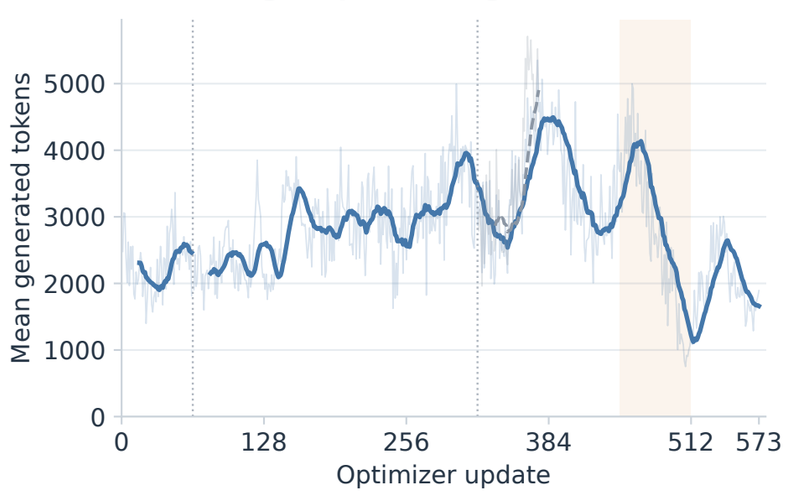
D Train / rollout mismatch



E Probability-ratio concentration



F Training response length



Retained lineage: pilot 1–64 → first continuation 65–320 → resumed continuation 321 onward. The failed attempt's updates 321–375 were discarded and repeated.
Update diagnostics first exist at step 1. W&B is missing pilot update 64; gaps are left unfilled. The plotted history contains 572 retained update records.
Mismatch compares Megatron training with vLLM rollout probabilities, not the original model. ESS describes token probability ratios, not independent tasks.